Zirong Chen

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EDUCATION

Vanderbilt University, Nashville, TN — Sept 2021 - May 2026

- *Doctor of Philosophy (Ph.D.)* in Computer Science; Advised by Dr. Meiyi Ma
- *Research Interest*: Artificial Intelligence, Natural Language Processing, Large Language Models
- *Dissertation Thesis*: Towards Trustworthy NLP-powered Intelligent Systems for Public Safety

Georgetown University, Washington, DC — Sept 2019 - May 2021

- *Master of Science (M.S.)* in Computer Science
- *Core Courses*: Deep Learning and Neural Nets, Empirical Methods in NLP, Deep Reinforcement Learning

Jinling Institute of Technology, Nanjing, China — Sept 2015 - May 2019

- *Bachelor in Engineering (B.E.)* in Computer Science and Technology
- *Thesis*: Security-Driven Big Data Applications in Smart Cities (collaborated with Nanjing Bureau of Public Security)

SELECTED AWARDS AND AWARDS

IJCAI Student Scholarship Awardee — June 2025
OpenAI Research Access Program Awardee ($10,000) — Nov 2024
AAAI Student Scholarship Awardee — Feb 2024
Vanderbilt Student Travel Grant Awardee (x3) — 2023; 2024; 2025

PUBLICATIONS

IJCAI-2025 **Zirong Chen**, Ziyan An, Jennifer Reynolds, Kristin Mullen, Stephen Martini, Meiyi Ma. *LogiDebrief: A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Models Integration*. Accepted at IJCAI-2025 (**13%**), Montreal, Canada.

AAAI-2025 **Zirong Chen**, Elizabeth Chason, Noah Mladenovski, Erin Wilson, Kristin Mullen, Stephen Martini, Meiyi Ma. *Sim911: Towards Effective and Equitable 9-1-1 Dispatcher Training with an LLM-Enabled Simulation*. Published (**18.9%**) and **Selected Oral Spotlight Presentation (4.9%)** at AAAI-2025, Philadelphia, PA.

AAAI-2024 **Zirong Chen**, Xutong Sun, Yuanhe Li, Meiyi Ma. *Auto311: A Confidence-Guided Automated System for Non-emergency Calls*. Published (**20.1**%) and **Selected Oral Spotlight Presentation (6%)** at AAAI-2024, Vancouver, Canada.

Pervasive and Mobile Computing **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec with shield: A secure intelligent assistant for requirement formalization*. Invited journal extension at Elsevier's Pervasive and Mobile Computing (2022)

IEEE-SmartComp-2022 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec: An intelligent assistant system for requirement specification in smart cities*. Published and **Best Paper Finalist** at IEEE SmartComp 2022, Helsinki, Finland.

AAMAS-2022 Ziyan An, Xia Wang, Hendrick Baier, **Zirong Chen**, Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. *Combing LLMs with a Logic-Based Framework to Explain MCTS*. Published (39.9%) at AAMAS-2025, Detroit, MI.

(Submitted to) **AAAI-2026** Ziyan An, Xia Wang, Hendrick Baier, **Zirong Chen**, Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. *LogiEx: Integrating Formal Logic and Large Language Model for Explainable Planning*. Submitted to AAAI-2026, Singapore, Singapore.

(Submitted to) **AAAI-2026** Siwoo Bae, Hanchen Wang, **Zirong Chen**, Meiyi Ma. *Learning with Preserving for Continual Multitask Learning*. Submitted to AAAI-2026, Singapore, Singapore.

(Submitted to) **AAAI-2026** **Zirong Chen**, Stephen Martini, Meiyi Ma. *A Hitchhiker's Guide to AI Adoption in High-Stake Training for Public Safety*. Submitted to AAAI-2026, Singapore, Singapore.

PROJECTS

Angie: AI-Driven Training System for 911 Call-takers in Emergency Responses Dec 2023 - Present
Integrating AI with real-world applications for a more efficient and effective emergency response training solution

- Collaborative Research Project with Metro Nashville Department of Emergency Communications, Metro Nashville Information Technology Services, and Metro Nashville Government. Funded by *OpenAI*, *Google*, *NSF(ReDDDoT)*, and *Vanderbilt*.
- Leading the project as a **major contributor** with local government agencies as AI tech vendor to work on a pioneering solution to transform emergency response systems by introducing AI in a responsible, reliable, and rigorous manner. This project includes sub-systems: *LogiDebrief*, *Sim911*, and *Auto311*. By far, Angie has been installed on over 400 workstations at local emergency communication center, has over 200 active users, and has been serving in Metro Nashville across different service cites for over 14 months.

LogiDebrief: An STL-based Automated 911 Call Debriefing System with LLMs Nov 2024 - Present
Disentangle overly complicated prompts into logic-enforced specifications with modularized use of LLMs

- Used signal temporal logic (STL) with LLMs to rigorously analyze and debrief past 911/311 calls. Avoid accuracy drop shown in traditional few-shot CoT + RAG frameworks as prompts get excessively long.
- Tested with real-world data, LogiDebrief reaches over 95% accuracy and saves nearly 11 mins per call for debriefing process.
- Successfully deployed at local emergency response center. It has been implemented into both daily use and new call-taking training programs. It has participated in automating over 1,701 emergency calls, saving over 311 work hours.

Sim911: An LLMs-based Simulation System for Call-taker Training Mar 2024 - Present
An effective and reliable workflow for LLMs role-play simulation under emergency responses

- Used CoT, RAG, few-shot prompting, designed and developed a context-aware controlled generation + looped validation and correction structure with LLM agents to ensure a realistic, true-to-life, and inclusive (considering needs from underserved groups) simulation experience. This framework is also under automatic correction, ensuring an improving performance under human feedback.
- Tested under real-world training programs, over 90% participants rate Sim911 as similar as or even better than human-led simulation experiences. The framework outperforms baseline by nearly 62% while benchmarking authenticity.
- Successfully deployed at local emergency response center. It has been implemented into new call-taker training programs. It has provided over 600 simulated calls, logging over 40 hours of active engagement, saving over 80 hours for trainers and instructors.

Auto311: A Confidence-Guided Automated System for Non-emergency Calls Jan 2023 - Present
Integrating AI with real-world applications for an automated call handling system

- Collaborated with Metro Nashville Department of Emergency Communications, Metro Nashville Information Technology Services, and Metro Nashville Government. Funded by NSF(CIVIC) and Vanderbilt.
- Designed and developed an automatic call handling system for non-emergency calls based on confidence measurement with deep neural networks. Outperformed naïve LLMs calls with 94.49% acc under emulation. Implemented baseline models like Transformers.

CitySpec: Converting Natural Languages to Formal Specifications Aug 2021 - Dec 2022
Rigorously and reliably translating natural languages to formal specifications

- Led the design and development of CitySpec and its extended version with shield, an end-to-end NLP tool for translating natural language descriptions into formal specifications, targeting urban planning applications.
- Utilized advanced semantic parsing techniques, like Named Entity Recognition (NER) and Part-of-Speech Tagging (PoS), and domain-specific datasets to enable accurate and scalable language-to-code translation.

INTERNSHIP EXPERIENCEs

Research Fellow, Epiq AI Labs (formerly LAER AI) – New York, NY May 2025 – Aug 2025

- Empowered AI Discovery Assistant^{TM} with **DeepResearch** capacities by agentic workflow design, especially under eDiscovery, document review, and other legal scenarios.
- Developed **FABRIC** (**F**ramework for **A**gent **B**enchmarking with **R**easoning and **I**nference **C**apacities) benchmark

with iterative bottom-up & top-down concept reconstruction and document generation. This benchmark provides tricky documents that contain fluff evidences and reasoning dead ends.

- Developed **LINT** (**L**earning via **IN**terative **T**utoring) testing environment with simulated user agents. This environment simulates user interactions and scores the agent-under-test performance during and after conversations. Simulated conversations can be restored anytime to rebuild any past conversation, allowing agent-under-test to iteractively improve. User agents follow the Blief-Desire-Intention (BDI) design and have multi-agent designs underlying to reflect human-like behaviors. Conversational behaviors and decisions are under sophisticated control from probabilistic finite state machines and persona configurations.

Data Scientist(NDA) , **National Basketball Association** (NBA) – New York, NY May - Aug 2020, 2021

- Analyzed user feedback with neural networks (PyTorch, Keras) and NLP tools (NLTK, spaCy) to improve fan engagement strategies.
- Optimized marketing campaigns by applying statistical machine learning models (scikit-learn) to enhance user experience and advertisement effectiveness.
- Built in-game data analysis pipelines using sequential neural networks (RNNs, Time2Vec) to predict streaming bandwidth requirements to further enhance profits.
- Designed scalable neural network architectures (PyTorch, Keras) for diverse applications, improving adaptability across projects.

ML Engineer(NDA) , **Sugon, Chinese Academy of Sciences –** Nanjing, China Dec 2018 - May 2019

- Developed projection algorithms based on vectorized path trajectories and distance metrics (Fréchet, Manhattan) to improve GPS accuracy in dense urban areas.
- Created dynamic routing algorithms using hidden Markov models to optimize urban planning and navigation tasks.
- Contributed to a high-impact, data-driven project involving predictive modeling and real-time decision-making for urban management applications.

INVITED TALKS AND SERVICES

AI for Education Demo at LIVE Learning Innovation Incubator	Mar 2024, 2025
Guest lectures at Vanderbilt (AI in CPS): “Large Language Models”	Feb 2023, 2024; Oct 2025
Guest lectures at Vanderbilt (AI): “Natural Language Processing”	Oct 2022, 2023, 2024
Guest lecture at Vanderbilt (Automated Verification): “Intro to Formal Logic”	Feb 2024
Keynote at NBA: “Autoencoders in Practice”	May 2021
Workshop at NBA: “Deep Learning: A Quick Start”	May 2020
Conference/Journal Program Committee (e.g., ICLR, AAAI, ACM Trans, etc.)	
Volunteer at AAAI-2024, IJCAI-2025	